

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12386035
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Schmid; Christian et al.

Semiconductor chip and device and method for driving at least one channel for a radar signal

Abstract

A semiconductor chip comprising at least one transmit channel and/or at least one receive channel for radar signals and also a sequencing circuit is proposed. In this case, the sequencing circuit is configured centrally to determine a sequencing scheme for time-dependent functions of the transmit channel and/or of the receive channel and to drive circuit elements of the transmit channel and/or of the receive channel in accordance with the sequencing scheme.

Inventors: Schmid; Christian (Linz, AT), Findenig; Rainer (Linz, AT), Greslehner-Nimmervoll; Bernhard (Hagenberg, AT)

Applicant: Infineon Technologies AG (Neubiberg, DE)

Family ID: 67400020

Assignee: Infineon Technologies AG (Neubiberg, DE)

Appl. No.: 17/819002

Filed: August 11, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220381880 A1	Dec. 01, 2022

Foreign Application Priority Data

DE	102018103088.6	Feb. 12, 2018
----	----------------	---------------

Related U.S. Application Data

continuation parent-doc US 16262987 20190131 US 11428779 child-doc US 17819002

Publication Classification

Int. Cl.: G01S7/40 (20060101); G01S7/03 (20060101); G01S7/35 (20060101); G01S13/87 (20060101)

U.S. Cl.:

CPC G01S7/4004 (20130101); G01S7/03 (20130101); G01S7/35 (20130101); G01S13/87 (20130101); G01S7/4056 (20130101)

Field of Classification Search

CPC: G01S (7/4004); G01S (7/03); G01S (7/35); G01S (13/87); G01S (7/4056); G01S (7/0232); G01S (7/0235); G01S (13/343); G01S (13/931); G01S (13/003); G01S (13/825); G01S (13/42); G01S (13/282); G01S (7/4911); G01S (13/345); G01S (13/34)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4540985	12/1984	Clancy et al.	N/A	N/A
6177904	12/2000	Coenen et al.	N/A	N/A
8513989	12/2012	Martin et al.	N/A	N/A
10509104	12/2018	Dato	N/A	G01S 13/87
10530611	12/2019	Manglani et al.	N/A	N/A
11035928	12/2020	Subburaj et al.	N/A	N/A
11428779	12/2021	Schmid et al.	N/A	N/A
2007/0178946	12/2006	Palum et al.	N/A	N/A
2014/0049423	12/2013	De et al.	N/A	N/A
2015/0002328	12/2014	Vaucher et al.	N/A	N/A
2015/0074293	12/2014	Nagata	N/A	N/A
2015/0153445	12/2014	Jansen	N/A	N/A
2016/0018511	12/2015	Nayyar et al.	N/A	N/A
2016/0025844	12/2015	McKitterick et al.	N/A	N/A
2016/0352388	12/2015	Lane	N/A	N/A
2017/0033919	12/2016	Sachse et al.	N/A	N/A
2017/0090015	12/2016	Breen et al.	N/A	N/A
2017/0139036	12/2016	Nayyar et al.	N/A	N/A
2018/0024233	12/2017	Searcy et al.	N/A	N/A
2018/0115409	12/2017	Nayyar	N/A	G01S 13/343

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101977021	12/2010	CN	N/A
102435981	12/2011	CN	N/A
104251984	12/2013	CN	N/A
104793189	12/2014	CN	N/A
106537170	12/2016	CN	N/A
106574967	12/2016	CN	N/A

OTHER PUBLICATIONS

Youxun, K., et al., "Radar Signal Generation System Based on DDS technology," 2006. cited by applicant

Koh, K.J., and Rebeiz, G.M., "A Q-Band Phased-Array Front-End with Integrated Wilkinson Couplers for Linear Power Combining in SiGe BiCMOS," IEEE BCTM 11.4, 2008, 4 pages. cited by applicant

Pang, J., "Implementation of Channelization and Digital Beamforming of Wideband Radar on FPGA," A thesis submitted to Xidian University, Dec. 2014, 82 pages. cited by applicant

Primary Examiner: Akonai; Olumide Ajibade

Attorney, Agent or Firm: Harrity & Harrity, LLP

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/262,987, filed Jan. 31, 2019 (now U.S. Pat. No. 11,428,779), which claims priority under 35 U.S.C. § 119 to German (DE) Patent Application No. 102018103088.6, filed on Feb. 12, 2018, the contents of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

(1) Example implementations are concerned with the control of radar systems. Example implementations are in particular concerned with a semiconductor chip and a device and a method for driving at least one channel for a radar signal.

BACKGROUND

(2) The integration level and complexity of radar sensors (e.g. in the automotive field) are increasing ever further. A multiplicity of functions within the radar sensor must be synchronized or operated in a synchronized manner.

(3) Frequency-modulated continuous wave radar (FMCW) systems are based on the transmission of frequency sweeps, also called ramps, which are defined by their start frequency, their stop frequency and their duration. Besides frequency generation itself, it is usually necessary to control further functions such as e.g. transmit channels, analog-to-digital converters (ADCs), monitoring functions or power amplifiers (PAs) synchronously with the frequency ramp.

(4) Complexity as well as power and/or resource consumption of the function synchronization should be as low as possible in this case.

SUMMARY

(5) There is thus a need to provide an improved technology for synchronizing functions in radar sensors or radar systems.

(6) The need can be met by the subject matter of the patent claims.

(7) A first example implementation relates to a semiconductor chip comprising at least one transmit channel and/or at least one receive channel for radar signals and also a sequencing circuit. In this case, the sequencing circuit is configured centrally to determine a sequencing scheme for time-dependent functions of the transmit channel and/or of the receive channel and to drive circuit elements of the transmit channel and/or of the receive channel in accordance with the sequencing scheme.

(8) A second example implementation relates to a system comprising a first semiconductor chip in accordance with one of the example implementations and a second semiconductor chip. In this

case, the sequencing circuit of the first semiconductor chip is configured to generate a control signal for a sequencing circuit of the second semiconductor chip. The control signal comprises at least one instruction for the sequencing circuit of the second semiconductor chip relative to a second sequencing scheme for time-dependent functions of at least one transmit channel and/or at least one receive channel for radar signals of the second semiconductor chip.

(9) A third example implementation relates to a system comprising a first semiconductor chip in accordance with one of the example implementations and a second semiconductor chip. The sequencing circuit of the first semiconductor chip is configured to generate a trigger signal for a sequencing circuit of the second semiconductor chip. The trigger signal causes the sequencing circuit of the second semiconductor chip to determine a sequencing scheme for time-dependent functions of at least one transmit channel and/or at least one receive channel of the second semiconductor chip and to drive circuit elements of the transmit channel and/or of the receive channel of the second semiconductor chip in accordance with the sequencing scheme.

(10) A fourth example implementation relates to a system comprising a first semiconductor chip in accordance with one of the example implementations and a second semiconductor chip having a memory and also at least one further transmit channel and/or at least one further receive channel for radar signals. A further sequencing scheme for time-dependent functions of the further transmit channel and/or of the further receive channel is stored in the memory. The control circuit of the first semiconductor chip is configured to generate a trigger signal that causes circuit elements of the further transmit channel and/or of the further receive channel to perform the time-dependent functions in accordance with the further sequencing scheme.

(11) A fifth example implementation relates to a device for driving at least one transmit channel for a radar signal. The device comprises a plurality of registers, wherein configuration parameters for circuit components of the transmit channel and frequency parameters for a respective assigned signal section of the radar signal are stored in the plurality of registers. Furthermore, the device comprises a control circuit configured to read the plurality of registers sequentially and to drive the circuit components of the transmit channel in each case on the basis of the information items stored in the plurality of registers.

(12) A sixth example implementation relates to a method for driving at least one transmit channel for a radar signal. The method comprises sequentially reading a plurality of registers, wherein configuration parameters for circuit components of the transmit channel and frequency parameters for a respective assigned signal section of the radar signal are stored in the plurality of registers. Furthermore, the method comprises driving the circuit components of the transmit channel in each case on the basis of the information items stored in the plurality of registers.

(13) Further example implementations additionally relate to a program comprising a program code for carrying out the method described herein if the program is executed on a processor or a programmable hardware component.

(14) Moreover, example implementations relate to a nonvolatile machine-readable medium on which is stored a program comprising a program code for carrying out the method described herein if the program is executed on a processor or a programmable hardware component.

(15) Furthermore, one example implementation relates to yet another device for driving at least one transmit channel for a radar signal. The device comprises means for sequentially reading a plurality of registers, wherein configuration parameters for circuit components of the transmit channel and frequency parameters for a respective assigned signal section of the radar signal are stored in the plurality of registers. Furthermore, the device comprises means for driving the circuit components of the transmit channel in each case on the basis of the information items stored in the plurality of registers.

BRIEF DESCRIPTION OF THE FIGURES

- (1) Some examples of devices and/or methods are explained in greater detail merely by way of example below with reference to the accompanying figures, in which:
- (2) FIG. 1 shows one example implementation of a semiconductor chip;
- (3) FIG. 2 shows one example implementation of a system comprising a plurality of semiconductor chips;
- (4) FIG. 3 shows a further example implementation of a system comprising a plurality of semiconductor chips;
- (5) FIG. 4 shows one example implementation of a device for driving at least one transmit channel for a radar signal;
- (6) FIG. 5 shows a further example implementation of a device for driving at least one transmit channel for a radar signal;
- (7) FIG. 6 shows one example implementation of a frequency ramp scenario;
- (8) FIG. 7 shows one example implementation of a sequence of instructions;
- (9) FIG. 8 shows a further example implementation of a device for driving at least one transmit channel for a radar signal;
- (10) FIG. 9 (presented as a combination of FIG. 9-1, FIG. 9-2, and FIG. 9-3) shows one example implementation of an instruction;
- (11) FIG. 10 shows a further example implementation of an instruction; and
- (12) FIG. 11 shows a flow diagram of one example implementation of a method for driving at least one transmit channel for a radar signal.

DESCRIPTION

- (13) Various examples will now be described thoroughly with reference to the accompanying figures, in which some examples are illustrated. In the figures, the thicknesses of lines, layers and/or regions may be illustrated in enlarged fashion for elucidation purposes.
- (14) While further examples are suitable for various modifications and alternative forms, some specific examples thereof are accordingly shown in the figures and described thoroughly below. However, this detailed description does not limit further examples to the specific forms described. Further examples can cover all modifications, counterparts and alternatives that fall within the scope of the disclosure. Throughout the description of the figures, identical reference signs refer to identical or similar elements which can be implemented identically or in modified form in a comparison with one another, while they provide the same or a similar function.
- (15) It goes without saying that if one element is designated as “connected” or “coupled” to another element, the elements can be connected or coupled directly or via one or more intermediate elements. If two elements A and B are combined using an “or”, this should be understood such that all possible combinations are disclosed, i.e. only A, only B, and A and B. An alternative wording for the same combinations is “at least one from A and B”. The same applies to combinations of more than two elements.
- (16) The terminology used here for describing specific examples is not intended to be limiting for further examples. If a singular form, e.g. “a, an” and “the”, is used and the use of only a single element is defined neither explicitly nor implicitly as obligatory, further examples can also use plural elements in order to implement the same function. If a function is described below as being implemented using a plurality of elements, further examples can implement the same function using a single element or a single processing entity. Furthermore, it goes without saying that the terms “comprises”, “comprising”, “has” and/or “having” in their usage indicate with greater precision the presence of the specified features, integers, steps, operations, processes, elements, components and/or a group thereof, but do not exclude the presence or the addition of one or more other features, integers, steps, operations, processes, elements, components and/or a group thereof.
- (17) Unless defined otherwise, all terms (including technical and scientific terms) are used here in

their customary meaning in the field with which examples are associated.

(18) FIG. 1 shows a semiconductor chip **100** comprising at least one transmit channel **110** for radar signals and/or at least one receive channel **120** for radar signals. The semiconductor chip **100** can thus be regarded as part of a radar sensor or part of a radar system. The transmit channel **110** comprises one or more circuit components and is configured to generate one or more radar transmission signals and to output the latter to one or more antennas. As indicated in FIG. 1, the transmit channel **110** can comprise e.g. one or more transmitters **111**. The receive channel **120** comprises one or more circuit components and is configured to receive and process one or more radar reception signals from one or more antennas. As indicated in FIG. 1, the receive channel **120** can comprise e.g. one or more receivers **121**, an analog front-end **122**, an ADC **123**, a digital front-end **124** and an interface **125**.

(19) Furthermore, semiconductor chip **100** comprises a sequencing circuit **130** configured centrally to determine a sequencing scheme for (e.g. all) time-dependent functions of the transmit channel **110** and/or of the receive channel **120** and also to drive circuit elements of the transmit channel **110** and/or of the receive channel **120** in accordance with the sequencing scheme. A time-dependent function of the transmit channel **110** and/or of the receive channel **120** is a function which is to be performed or carried out in a manner temporally coordinated or synchronized with other time-dependent functions of the transmit channel **110** and/or of the receive channel **120** in order to ensure correct operation of the radar sensor or radar system. Accordingly, the sequencing scheme represents the temporally coordinated or synchronized order of performance of the individual time-dependent functions.

(20) By way of example, the time-dependent functions can comprise a function of the transmit channel **110** relating to generating a radio-frequency radar transmission signal, a function of the receive channel **120** relating to processing a radio-frequency radar reception signal, a monitoring function for one or more circuit components of the transmit channel **110** and/or of the receive channel **120** and/or a monitoring function for a signal processed by the transmit channel **110** and/or the receive channel **120**.

(21) The sequencing circuit **130** can be implemented for example as a dedicated circuit or a circuit for executing software in conjunction with associated implemented software configured to determine a sequencing scheme and to drive circuit elements of the transmit channel **110** and/or of the receive channel **120** in accordance with the sequencing scheme.

(22) Radar operation of the at least one transmit channel **110** situated on the semiconductor chip **100** and/or of the at least one receive channel **120** situated on the semiconductor chip **100** is controlled centrally by the sequencing circuit **130** arranged on the semiconductor chip **100**. Accordingly, the radar operation on the semiconductor chip **100** can be effected substantially autonomously, i.e. independently of external controllers, on account of the sequencing circuit **130**.

(23) The semiconductor chip **100** can be used for example for a (phase- or frequency-) modulated continuous wave radar system. Accordingly, the transmit channel **110** and/or the receive channel **120** can be part of a modulated continuous wave radar system, for example of an FMCW radar in the automotive field. The sequencing circuit **130** can accordingly be understood as a central sequencing unit which can coordinate all time-critical functions of the automotive FMCW radar front-end, such that a radar operation is an autonomous process which for example does not necessitate the participation of an external processor. In comparison with systems in which the synchronization of functions is distributed via a plurality of chips (e.g. a central microcontroller chip for controlling and/or for reconfiguring monitoring functions and also a phase-locked loop chip for providing a desired frequency profile), the development outlay can be lower. On account of the autonomous radar operation of the semiconductor chip **100**, a load for a main processor of a radar sensor or of a radar system can be reduced and a higher flexibility can be achieved. Furthermore, a very high synchronism (e.g. greater than in customary processors) can be made possible on account of the possibility for the specific design of the sequencing unit.

(24) In addition to e.g. controlling a desired frequency profile of a radar transmission signal, the sequencing circuit **130** shown in FIG. **1** can control various other aspects of a radar sensor or radar system in a synchronized manner. By way of example, in the transmitter **111**, a PA can be switched on and off in a synchronized manner or a phase shift of radar transmission signals can be carried out in a synchronized manner. The monitoring of the transmit channel **110** and/or of the receive channel **120** can be carried out e.g. by means of a monitoring circuit **140**. The latter can accordingly be driven by the sequencing circuit **130** to trigger or to activate or to deactivate a monitoring function in a synchronized manner in accordance with the sequencing scheme. By way of example, monitoring functions can be activated or deactivated in a synchronized manner during ramp formation, a calibration (e.g. amplification of a voltage-controlled oscillator), a cascaded operation or a configuration of external components or slaves (e.g. Serial Peripheral Interface (SPI) or DeMULTipleXing (DMUX)).

(25) In the signal generating circuit **150**, in a synchronized manner, for example, the bandwidth of a phase-locked loop (PLL) can be set (e.g. charge pump current) or more extensive modulation concepts can be activated or deactivated (e.g. 2-point modulation, resetting current). Likewise, in a synchronized manner, for example, loop filter reset circuits can be activated or deactivated or monitoring functions can be activated or deactivated. Moreover, in a synchronized manner, by way of example, voltages can be roughly set, calibration parameters (e.g. offset current, anti-backlash, amplification of a voltage-controlled oscillator) can be set or a signal source (e.g. when using a plurality of PLLs) can be selected.

(26) In the receiver **121**, in a synchronized manner, by way of example, digital or analog filters can be set (resetting, configuration, bypass . . .) or decimation rates can be set. LVDS (Low Voltage Differential Signaling) modes or calibrations can be chosen in a synchronized manner. Likewise, in a synchronized manner, the receive channel **120** can e.g. also be activated or deactivated in its entirety or a reception frame delay can be set. By way of example, the ADC **123** can also be configured, calibrated, activated or deactivated in a synchronized manner.

(27) Likewise, general purpose inputs/outputs (GPIOs) can be configured in a synchronized manner, circuit components for generating frequency ramps can be triggered or diverse circuit components (e.g. ADC **123**) can be switched on or off. Moreover, interrupts can be generated or other chip functionality can be triggered.

(28) In order to synchronize the time-dependent functions presented by way of example above, the sequencing circuit **130** can comprise an instruction-processing unit (not shown) with a specific instruction set for determining the sequencing scheme. The instruction set can describe the configuration flow over time and be for example similar to the instruction set of a general purpose processor. By way of example, the instruction set can comprise first specific instructions for the configuration of the circuit elements of the transmit channel **110** and/or of the receive channel **120** and also second specific instructions for the configuration of frequency parameters of a radio-frequency radar transmission signal generated by the transmit channel **110**. Details of the specific instruction set are explained more specifically below with reference to FIGS. **4** to **10**.

(29) Sequencing circuit **130** can thus be understood as a radar-specific sequencing unit having an instruction set associated with a specific purpose in order to handle time-critical configurations in a highly integrated radar chip.

(30) The master/slave interface **160** shown in FIG. **1** can additionally be used to control or supervise further radar chips in a cascading of a plurality of chips. Some possible implementations are described below with reference to FIGS. **2** and **3**.

(31) FIG. **2** shows system **200** comprising a first semiconductor chip **210** in accordance with the principles described in association with FIG. **1** and also a second semiconductor chip **220-1** and optionally further second semiconductor chips **220-n**. The first semiconductor chip **210** is embodied as a master, wherein the second semiconductor chip **220-1** and also the further second semiconductor chips **220-n** are embodied as slaves.

(32) The sequencing circuit **211** of the first semiconductor chip **210** is configured to generate a control signal for a sequencing circuit **221-1** of the second semiconductor chip **220-1** (and correspondingly optionally also for the sequencing circuits **221-n** of the further second semiconductor chips **220-n**). The control signal is output via the master/slave interface **212** of the first semiconductor chip **210** to the master/slave interface **222-1** of the second semiconductor chip **220-1**. The control signal comprises at least one instruction for the sequencing circuit **221-1** of the second semiconductor chip **220-1** relative to a second sequencing scheme for time-dependent functions of at least one transmit channel (not shown) and/or of at least one receive channel (not shown) for radar signals of the second semiconductor chip **220-1**. The sequencing circuit **211** of the first semiconductor chip can thus determine instructions for the second semiconductor chip **220-1** and forward them to the latter in order that the transmit channel and/or the receive channel of the second semiconductor chip **220-1** are/is correspondingly driven by the sequencing circuit **221-1** of the second semiconductor chip **220-1**. In other words: the master/slave interface **212** of the first semiconductor chip **210** can transfer sequence instructions to the slave unit, such that the sequencing unit in the master chip can transfer instructions to the slave unit.

(33) Alternatively, the sequencing circuit **211** of the first semiconductor chip **210** can also be configured to generate a trigger signal for the sequencing circuit **221-1** of the second semiconductor chip **220-1** (and correspondingly optionally also for the sequencing circuits **221-n** of the further second semiconductor chips **220-n**). The trigger signal causes the sequencing circuit **221-1** of the second semiconductor chip **220-1** to determine a sequencing scheme for time-dependent functions of at least the transmit channel and/or the receive channel of the second semiconductor chip **220-1** and to drive circuit elements of the transmit channel and/or of the receive channel of the second semiconductor chip **220-1** in accordance with the sequencing scheme. The sequencing circuit **211** of the first semiconductor chip can thus trigger the synchronization and/or sequencing functionality of the second semiconductor chip **220-1**. In other words: the master/slave interface **212** of the first semiconductor chip **210** can transfer trigger signals to the slave unit in order to trigger a sequencing unit in the slave chip.

(34) FIG. 3 shows system **300** comprising a first semiconductor chip **310** in accordance with the principles described in association with FIG. 1 and also a second semiconductor chip **320-1** and optionally further second semiconductor chips **320-n**. The first semiconductor chip **310** is embodied as master, wherein the second semiconductor chip **320-1** and also the further second semiconductor chips **320-n** are embodied as slaves.

(35) The second semiconductor chip **320-1** comprises a memory **311-1** and also at least one transmit channel (not shown) and/or at least one receive channel (not shown) for radar signals. A further sequencing scheme for time-dependent functions of the further transmit channel of the second semiconductor chip **320-1** and/or of the further receive channel of the second semiconductor chip **320-1** is stored in the memory **311-1**. Accordingly, the further second semiconductor chips **320-n** also each comprise a memory **311-n** and at least one transmit channel (not shown) and/or at least one receive channel (not shown) for radar signals.

(36) The control circuit **311** of the first semiconductor chip **310** is configured to generate a trigger signal that causes circuit elements of the further transmit channel and/or of the further receive channel of the second semiconductor chip **320-1** (and correspondingly optionally of the further second semiconductor chips **320-n**) to perform the time-dependent functions in accordance with the further sequencing scheme stored in the memory **321-1** (and respectively the memories **321-n**). In other words: only the master chip, but not the slave chips, uses a sequencing unit, wherein the sequencing unit can instruct the slave chips directly to apply the sequencing scheme stored in them.

(37) Example driving of a transmit channel for a radar signal is described below with reference to FIGS. 4 to 10.

(38) In this case, FIG. 4 shows a device **400** for driving at least one transmit channel **430** for a radar signal. Device **400** can be e.g. part of the sequencing circuit **130** shown in FIG. 1.

(39) Device **400** comprises a plurality of registers **410-1, 410-2, 410-3, 410-4 . . . 410-n**.

Configuration parameters for circuit components of the transmit channel **430** and frequency parameters for a respective assigned signal section of the radar signal are stored in the plurality of registers **410-1, 410-2, 410-3, 410-4 . . . 410-n**.

(40) The configuration parameters for the circuit components of the transmit channel **430** comprise specific configurations for one or more circuit components of the transmit channel (e.g. an ADC, a PA or a monitoring function). The radar signal includes a plurality of successive signal sections. The frequency parameters for the respective signal section indicate e.g. a start frequency, a stop frequency and/or a duration of the signal section. By way of example, a signal section can correspond to a ramp segment of a frequency ramp of the radar signal (as shown in FIG. 6).

(41) Furthermore, device **400** comprises a control circuit **420** configured to read the plurality of registers sequentially **410-1, 410-2, 410-3, 410-4 . . . 410-n** and to drive the circuit components of the transmit channel **430** in each case on the basis of the information items stored in the plurality of registers.

(42) In this way, the radar signal can be generated segment by segment on the basis of the sequentially read plurality of registers sequentially **410-1, 410-2, 410-3, 410-4 . . . 410-n**.

(43) For a single signal section of the radar signal, the configuration parameters for the circuit components of the transmit channel **430** and the frequency parameters for the assigned signal section can be stored either jointly in a single register or in separate registers.

(44) By way of example, the plurality of registers **410-1, 410-2, 410-3, 410-4 . . . 410-n** can comprise a first register set having a plurality of first registers and a second register set having a plurality of second registers. In this case, the configuration parameters are stored in the plurality of first registers, wherein the frequency parameters together with at least one reference to one of the first registers are stored in the plurality of second registers. In other words: instead of jointly storing the configuration parameters and the frequency parameters of a signal section of the radar signal (e.g. of a ramp segment), they can also be stored in separate registers.

(45) This is shown in FIG. 5. Device **500** comprises the first register set **510** having the plurality of first registers, in which the configuration parameters are stored, and the second register set **520** having the plurality of second registers, in which the frequency parameters for a respective associated signal section together with at least one reference to one of the first registers are stored.

(46) The first registers contain configuration parameters for (all) internal and external circuits or modules of the transmit channel which have to be controlled synchronously for the generation of the respective signal section of the radar signal (e.g. for generating a frequency ramp).

(47) Not only a start and stop frequency of a signal section (e.g. ramp segment) and the duration thereof, but also the reference to one or more first registers are stored in the second registers. This can make it possible that a specific configuration (in the form of the configuration parameters) can be programmed and stored only a single time, since reference can be made to this setting by way of the reference stored in the respective second register by virtue of a multiplicity of second registers. For various/different signal sections, reference can thus be made to the configuration stored in said one first register. The configuration does not have to be stored separately or anew for each signal section. Splitting the registers into the two register sets thus makes it possible to reuse configuration data programmed previously. The memory demand for generating the individual signal sections of the radar signal can thus be reduced by the splitting.

(48) Furthermore, a first loop information item indicating a beginning of a loop and a (finite or infinite) number of loop iterations can be stored in a first register of the plurality of registers (e.g. a first of the plurality of second registers **520**) and a second loop information item indicating an end of the loop can be stored in a second register of the plurality of registers (e.g. a second of the plurality of second registers **520**). The control circuit **420** is then correspondingly furthermore configured to read the registers starting from the first of the plurality of registers up to the second of the plurality of registers in order to generate a loop body and to drive the circuit components of

the transmit channel **430** repeatedly in accordance with the number of loop iterations on the basis of the loop body. This can enable a significant additional reduction of the memory demand. This is explained in greater detail below with reference to the frequency ramp scenario shown in FIG. 6.

(49) The frequency ramp scenario shown in FIG. 6 shows the temporal frequency profile of a radar signal and includes a plurality of ramp sequences **610, 620, 630, . . .**, which are identical with regard to their profile but frequency-shifted with respect to one another. Each of the ramp sequences **610, 620, 630, . . .** includes a plurality of frequency ramps each consisting of a ramp segment a, a ramp segment b and a ramp segment c. Each of the ramp segments a, b and c is determined by its start frequency, its stop frequency and its duration. The three b ramp segments in each ramp sequence can be generated e.g. using three transmit channels. Although just under thirty ramp segments are defined, four configurations can suffice to generate the radar signal. A respective configuration is used for each of the three activated transmit channels, and also an additional configuration in which all the transmit channels are deactivated. The amount of memory required for programming or storing the ramp scenario can thus be significantly reduced by the splitting of the registers.

(50) By means of a memory circuit (not shown in FIG. 4), on the basis of a received instruction, it is possible to update at least one configuration parameter in one or more of the plurality of registers **410-1, 410-2, 410-3, 410-4 . . . 410-n** and/or to update at least one frequency parameter in one or more of the plurality of registers **410-1, 410-2, 410-3, 410-4 . . . 410-n** and/or to update at least one loop information item in one or more of the plurality of registers **410-1, 410-2, 410-3, 410-4 . . . 410-n**.

(51) Likewise, a received instruction can comprise execution information items indicating whether one of the plurality of registers is intended to be executed. Accordingly, the control circuit **420** can furthermore be configured, depending on the execution information items, to drive the circuit components of the transmit channel **430** on the basis of the information items stored in said one of the plurality of registers **410-1, 410-2, 410-3, 410-4 . . . 410-n**.

(52) The instructions can be embodied as opcodes e.g. in a microprocessor-like or -similar approach.

(53) The programming of the ramp scenario illustrated in FIG. 6 can be carried out here with just ten instructions (e.g. opcodes). This is illustrated by way of example in FIG. 7.

(54) By means of the first four instructions **710, 720, 730** and **740**, the four configurations for the frequency segments a, b, c of each ramp sequence in the form of the configuration parameters are programmed or stored in the plurality of registers (e.g. in four registers of the plurality of first registers **510**). A respective configuration is stored for each of the three activated transmit channels in one of the registers, and also an additional configuration in which all the transmit channels are deactivated.

(55) The ramp segments a, b and c are manifested three times in each ramp sequence, wherein the ramp segments a and c refer to the register having the configuration in which all the transmit channels are deactivated, while the segment b iterates over registers having the configurations for the three transmit channels. That is to say that three instructions **750, 760** and **770** are used, which additionally program or store the loop information items in three (of the plurality of second) registers in each case besides the initial frequency parameters for the ramp segments a, b and c of the first ramp sequence and also the references to the associated registers (from among the plurality of first registers). The first ramp sequence **610** can be executed in this way.

(56) Afterward, another three instructions **780, 790** and **700** are used to shift the frequency parameters (e.g. the start frequencies) in the three (of the plurality of second) registers for the transition from one ramp sequence to the next (e.g. from the first ramp sequence **610** to the second ramp sequence **620**). The respective registers need not be executed or are not executed—just the register value is updated.

(57) Once all the register values have been updated, the outer loop jumps back, and the second

ramp sequence **620** is generated on the basis of the registers updated in accordance with the instructions **780**, **790** and **700**.

(58) The entire loop is repeated until the predetermined number of loop iterations has been reached.

(59) As indicated above, a plurality of references to different registers from among the plurality of first registers **510** can be stored at least in one of the second registers **520**. Accordingly, the control circuit **420** can then furthermore be configured, in the case of multiple driving of the circuit components of the transmit channel **430**, on the basis of said one second register, to use respectively a different one of the plurality of references to the different first registers for the driving of the circuit components of the transmit channel **430**. Relative to the above example in FIG. **6**, upon each execution of the second register assigned to the ramp segment b, a different one of the references to the three first registers in which the configuration parameters for one of the three transmit channels are respectively stored is used for the generation of the radar signal. That is to say that the second registers **520** can be designed not only to refer to an individual one of the first registers **510**, but to iterate over a range of same. This allows the configuration to be switched if one of the second registers **520** is reused, without requiring a separate instruction (e.g. opcode) for this purpose.

(60) FIG. **8** shows with a greater degree of detail a device **800** for driving at least one transmit channel for a radar signal. Device **800** comprises a plurality of registers, wherein configuration parameters for circuit components of the transmit channel and frequency parameters for a respective assigned signal section of the radar signal are stored in the plurality of registers. The plurality of registers comprises a first register set **810** having a plurality of first registers and a second register set **820** having a plurality of second registers. The configuration parameters are stored in the plurality of first registers **810**. The frequency parameters are stored together with at least one reference to one of the first registers in the plurality of second registers **820**.

(61) A control circuit **840** of the device **800** is configured to read the plurality of second registers **820** sequentially and, by means of the sequencing circuit **841**, to drive the circuit components of the transmit channel in each case on the basis of the information items stored in the plurality of second registers **820**.

(62) Furthermore, device **800** comprises a memory circuit **830** configured, on the basis of a received instruction, to update at least one configuration parameter in one or more of the plurality of (first) registers and/or to update at least one frequency parameter in one or more of the plurality of (second) registers. The memory circuit can be designed e.g. as a Direct Memory Access (DMA) engine that accesses an internal memory **850** of the transmit channel or of the radar system and reads the instructions (e.g. opcodes) from the latter in order to program or store the configuration and frequency parameters correspondingly in the plurality of registers. As indicated above, the instruction can comprise update information for one or more of the plurality of registers.

(63) By way of example, the received instruction can comprise items of information about an updated absolute value of the configuration parameter in said one of the plurality of (e.g. first) registers and/or items of information about an updated absolute value of the frequency parameter in said one of the plurality of (e.g. second) registers. Accordingly, the memory circuit **830** can be configured to overwrite an instantaneous value of the configuration parameter with the updated absolute value of the configuration parameter and/or to overwrite an instantaneous value of the frequency parameter with the updated absolute value of the frequency parameter. Likewise, the instruction can comprise updated loop information items indicating a beginning of a loop, an end of a loop and/or a number of loop iterations. Accordingly, the memory circuit **830** can be configured to overwrite instantaneous loop information items with the updated loop information items.

(64) Alternatively, the received instruction can also comprise items of information about a change in an instantaneous value of the frequency parameter in said one of the plurality of registers. Memory circuit **830** can then be configured to change the instantaneous value of the frequency parameter in accordance with the items of information about the change in the instantaneous value

of the frequency parameter.

(65) In other words: registers can be updated on the basis both of absolute and of relative update information.

(66) One example of an instruction **900** for updating one or more of the second registers **820** is shown in the form of an opcode in FIG. **9**. FIG. **9** is shown as a combination of three figures: FIG. **9-1**, FIG. **9-2**, and FIG. **9-3**. FIG. **9-1** is the left-most figure of FIG. **9**. FIG. **9-2** is the middle figure of FIG. **9**. FIG. **9-3** is the right-most figure of FIG. **9**.

(67) As shown in FIG. **9**, the instruction **900** comprises at least one header **910**, in which individual bits indicate what information is contained in the instruction. As indicated in FIG. **9**, the header **910** can indicate, for example, which or which of the plurality of second registers **820** ought to be programmed, whether loop information items ought to be stored in a register or which frequency parameters (e.g. start frequency, stop frequency, duration) ought to be updated. Depending on the bits set in the header **910**, the instruction **900** contains additional structures (words). Parts of the instruction (e.g. opcode) which are used for programming the registers can thus be omitted in order selectively to update only parts of the registers.

(68) As indicated in FIG. **9**, the instruction **900** can have e.g. in each case additional structures for the number of loop executions, the duration of the signal section assigned to the register, the start frequency, the frequency step and/or the reference to one or more of the first registers **810**. The fields for the duration, the start frequency and the stop frequency can contain e.g. a bit for indicating whether the new value should be added to the value currently stored in the segment register (i.e. relative change) or whether the latter ought to be overwritten (i.e. absolute change). The configuration reference field contains a bit for indicating whether it contains a single, fixed reference to one of the first registers **810**, or a range of first registers **810** over which the ramp segment ought to iterate in different executions. It is thus possible to omit parts of the instruction for programming a register if only parts thereof ought to be updated. Furthermore, the instruction can specify whether each part of the register ought to be set to an absolute value or whether a value ought to be modified relative to the value stored previously.

(69) The header **910** additionally contains the bit “Execute segment” for selecting whether the updated segment ought to be executed by the control circuit **840** or whether the instruction ought only to modify the register. Furthermore, the header **910** also contains the bit “End of sequence” for indicating that the ramp sequence is ended after this segment.

(70) In other words: a received instruction can furthermore comprise execution information items indicating whether one of the plurality of (second) registers is intended to be executed. The control circuit **840** is then configured, depending on the execution information items, to drive the circuit components of the transmit channel on the basis of the information items stored in said one of the plurality of (second) registers. Likewise, a received instruction can comprise termination information for one of the plurality of (second) registers. The control circuit **840** is then configured, on the basis of the termination information, to drive the circuit components of the transmit channel to end the generation of the radar signal after the signal section assigned to said one of the plurality of (second) registers.

(71) Using the loop bits in header **910**, loops can furthermore be defined—without any additional instructions. In the control circuit **840**, using a stack data structure, it is possible to store start address or start register and count of the loop in a separate, internal loop stack **842**. By way of example, if a segment or associated second register with the bit “Beginning of a new loop” is executed, its address and its loop count are pushed onto said loop stack **842**. After the execution of a segment or of the associated second register with the bit “End of current loop” (possibly the same as at the beginning of the loop), the topmost loop count of the loop stack **842** is decremented and, if the loop is not ended, the start of the loop is loaded as the next instruction (e.g. opcode).

(72) In other words: a first loop information item indicating a beginning of a loop and a number of loop iterations is stored in a first register of the plurality of second registers **820**, and a second loop

information item indicating an end of the loop is stored in a second register of the plurality of second registers **820**. The control circuit **840** is configured to read the second registers starting from the first of the plurality of second registers **820** up to the second of the plurality of second registers **820** in order to generate a loop body and to drive the circuit components of the transmit channel repeatedly in accordance with the number of loop iterations on the basis of the loop body.

(73) One example of an instruction **1000** for updating one or more of the first registers **810** is shown in the form of a further opcode in FIG. **10**. The instruction **1000** once again comprises a header **910**, in which individual bits indicate at which register of the plurality of first registers **810** the update ought to be begun and the number of registers to be updated. For each register to be changed, a high word and/or a low word with the new configuration for the register are/is additionally contained.

(74) The control circuit **840** can begin the driving of the transmit channel for generating the ramp sequence e.g. on account of an external trigger.

(75) While a segment or the associated register(s) is/are executed, all other registers not currently being used can be reprogrammed by further instructions (e.g. opcodes). By way of example, it is possible to copy the information items from the associated registers into the loop stack **842** in order to be executed or processed by the control circuit **840**. Even while the register information items stored in the loop stack **842** are being processed, the first registers **810** and/or the second registers **820** can be updated. Referring to the example shown in FIG. **7**, the frequency shift instructions **780**, **790** and **700** for the ramp segments a, b and c can already be executed while the segment c is still being processed by the control circuit. In this regard, the updated segment a can be executed directly after segment c has ended.

(76) The memory circuit **830** is synchronized with the control circuit **840** for this purpose, such that registers currently being used (e.g. read or copied into the loop stack **842**) by the control circuit **840** are not overwritten. By way of example, the memory circuit **830** can be stopped until the control circuit **840** has finished reading or executing the registers currently being used, in order subsequently to update them. If the register to be updated is for example a register which is still to be executed and which has not yet been used or processed by the control circuit (e.g. because the control circuit **840** is still busy processing information for previous ramp segments), in accordance with example implementations, the memory circuit **830** can be stopped until the register has been used by the control circuit **840**. This can allow e.g. new parameters to be loaded while others are being executed. This can make it possible to execute ramp sequences which as a whole are too large to fit into the plurality of registers.

(77) In other words: the memory circuit **830** can furthermore be configured to update one of the plurality of registers only if said one of the plurality of registers is not being instantaneously accessed by the control circuit **840**. Furthermore, the memory circuit is then configured to update said one of the plurality of registers as soon as the control circuit **840** is no longer accessing said one register.

(78) In order to summarize the above-described aspects for synchronizing functions in radar sensors or radar systems, FIG. **11** shows a flow diagram of a method **1100** for driving at least one transmit channel for a radar signal. Method **1100** comprises sequentially reading **1102** a plurality of registers, wherein configuration parameters for circuit components of the transmit channel and frequency parameters for a respective assigned signal section of the radar signal are stored in the plurality of registers. Furthermore, method **1100** comprises driving **1104** the circuit components of the transmit channel in each case on the basis of the information items stored in the plurality of registers.

(79) Further details and aspects of the method have been described above in association with further example implementations (e.g. FIGS. **4** to **10**). The method can comprise one or more optional features in accordance with the further example implementations.

(80) At the system level, a central sequencing or function synchronization in accordance with the

present concept eases the demand for a time-critical interaction between e.g. a master microcontroller and the radar chip. By additionally reducing the memory demand required for a ramp sequence, it is possible further to minimize the processing load for the master microcontroller. (81) The aspects and features that have been described together with one or more of the examples and figures described in detail above can also be combined with one or more of the other examples in order to replace an identical feature of the other example or in order additionally to introduce the feature into the other example.

(82) Examples can furthermore be or be related to a computer program comprising a program code for carrying out one or more of the above methods when the computer program is executed on a computer or processor. Steps, operations or processes of various methods described above can be performed by programmed computers or processors. Examples can also cover program storage devices, e.g. digital data storage media, which are machine-, processor- or computer-readable and code machine-executable, processor-executable or computer-executable programs of instructions. The instructions carry out some or all of the steps of the methods described above or cause them to be carried out. The program storage devices can comprise or be e.g. digital memories, magnetic storage media such as, for example, magnetic disks and magnetic tapes, hard disk drives or optically readable digital data storage media. Further examples can also cover computers, processors or control units which are programmed to carry out the steps of the methods described above, or (field) programmable logic arrays ((F)PLAs) or (field) programmable gate arrays ((F)PGAs) which are programmed to carry out the steps of the methods described above.

(83) The description and drawings present only the principles of the disclosure. Furthermore, all examples mentioned here are intended to be used expressly only for teaching purposes, in principle, in order to assist the reader in understanding the principles of the disclosure and the concepts contributed by the inventor(s) for further development of the art. All statements herein regarding principles, aspects and examples of the disclosure and also concrete examples thereof are intended to encompass the counterparts thereof.

(84) A functions block designated as “means for . . .” carrying out a specific function can relate to a circuit or software configured for carrying out a specific function. Consequently, a “means for something” can be implemented as a “means configured for or suitable for something”, e.g. a component, a circuit or software configured for or suitable for the respective task.

(85) Functions of different elements shown in the figures including those function blocks designated as “means”, “means for providing a signal”, “means for generating a signal”, etc. can be implemented in the form of dedicated hardware, e.g. “a signal provider”, “a signal processing unit”, “a processor”, “a controller”, etc., and as hardware capable of executing software in conjunction with associated software. When provided by a processor, the functions can be provided by a single dedicated processor, by a single jointly used processor or by a plurality of individual processors, some or all of which can be used jointly. However, the term “processor” or “controller” is far from being limited to hardware capable exclusively of executing software, but rather can encompass digital signal processor hardware (DSP hardware), application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), read only memory (ROM) for storing software, random access memory (RAM) and non-volatile memory devices (storage). Other hardware, conventional and/or customized, can also be included.

(86) A block diagram can illustrate for example a rough circuit diagram which implements the principles of the disclosure. In a similar manner, a flow diagram, a flow chart, a state transition diagram, a pseudo-code and the like can represent various processes, operations or steps which are represented for example substantially in a computer-readable medium and are thus performed by a computer or processor, regardless of whether such a computer or processor is explicitly shown. Methods disclosed in the description or in the patent claims can be implemented by a component having a means for performing each of the respective steps of said methods.

(87) It goes without saying that the disclosure of a plurality of steps, processes, operations or

functions disclosed in the description or the claims should not be interpreted as being in the specific order, unless this is explicitly or implicitly indicated otherwise, e.g. for technical reasons. The disclosure of a plurality of steps or functions therefore does not limit them to a specific order unless said steps or functions are not interchangeable for technical reasons. Furthermore, in some examples, an individual step, function, process or operation can include a plurality of partial steps, functions, processes or operations and/or be subdivided into them. Such partial steps can be included and be part of the disclosure of said individual step, provided that they are not explicitly excluded.

(88) Furthermore, the claims that follow are hereby incorporated in the detailed description, where each claim can be representative of a separate example by itself. While each claim can be representative of a separate example by itself, it should be taken into consideration that—although a dependent claim can refer in the claims to a specific combination with one or more other claims—other examples can also encompass a combination of the dependent claim with the subject matter of any other dependent or independent claim. Such combinations are explicitly proposed here, provided that no indication is given that a specific combination is not intended. Furthermore, features of a claim are also intended to be included for any other independent claim, even if this claim is not made directly dependent on the independent claim.

Claims

1. A semiconductor radar chip, comprising: a transmit channel configured to transmit a frequency modulated continuous wave radar signal comprising a plurality of frequency ramps; and a sequencing circuit configured to use loop instructions generating a number of loop iterations and frequency information to determine a sequencing scheme for time-dependent functions of the transmit channel and to drive circuit components of the transmit channel in accordance with the sequencing scheme, wherein the circuit components of the transmit channel are driven repeatedly in accordance with the number of loop iterations in order to generate the plurality of frequency ramps, and wherein each frequency ramp, of the plurality of frequency ramps, is assigned to a loop iteration of the number of loop iterations.
2. The semiconductor radar chip of claim 1, wherein a loop body is generated for each loop of the number of loop iterations, and wherein the circuit components of the transmit channel are driven repeatedly in accordance with the number of loop iterations based on the loop body.
3. The semiconductor radar chip of claim 1, wherein the frequency information is updated during an execution of the number of loop iterations.
4. The semiconductor radar chip of claim 3, wherein the frequency information is updated during the execution of the number of loop iterations by shifting a start frequency and stop frequency.
5. The semiconductor radar chip of claim 1, further comprising: a receive channel, wherein the sequencing circuit is configured to drive circuit elements of the receive channel in accordance with the sequencing scheme.
6. The semiconductor radar chip of claim 1, wherein the sequencing circuit is configured to use deactivation information to determine the sequencing scheme, wherein the deactivation information indicates that the transmit channel to be deactivated.
7. The semiconductor radar chip of claim 1, wherein the loop instructions are executed in an instruction-processing unit with a specific instruction set for determining the sequencing scheme.
8. The semiconductor radar chip of claim 1, wherein the semiconductor radar chip is configurable to generate a trigger signal to trigger a second sequencing circuit of a second semiconductor radar chip, wherein the second sequencing circuit of the second semiconductor radar chip is configured to generate a second number of loop iterations and second frequency information to determine a second sequencing scheme for second time-dependent functions of a second transmit channel of the second semiconductor radar chip and to drive second circuit elements of the second transmit

channel in accordance with the second sequencing scheme.

9. The semiconductor radar chip of claim 1, wherein the sequencing circuit comprises a plurality of registers including: a first register storing one or more configuration parameters for the circuit components, and a second register storing one or more frequency parameters for a signal section associated with the plurality of registers.

10. A system, comprising: a first semiconductor chip comprising: a transmit channel for processing a first radar signal; and a sequencing circuit configured to use loop instructions generating a number of loop iterations and frequency information to determine a sequencing scheme for time-dependent functions of the transmit channel and to drive circuit elements of the transmit channel in accordance with the sequencing scheme; and a second semiconductor chip comprising: a second transmit channel for processing a second radar signal; and a second sequencing circuit configured to receive a trigger signal from the first radar signal and generate a second number of loop iterations and second frequency information to determine a second sequencing scheme for second time-dependent functions of the second transmit channel and to drive second circuit elements of the second transmit channel in accordance with the second sequencing scheme.

11. The system of claim 10, wherein at least one of: the transmit channel is configured to transmit a frequency modulated continuous wave radar signal comprising a plurality of frequency ramps, the circuit elements of the transmit channel are driven repeatedly in accordance with the number of loop iterations in order to generate the plurality of frequency ramps, or the second transmit channel is configured to transmit a second frequency modulated continuous wave radar signal comprising a second plurality of frequency ramps, the second circuit elements of the second transmit channel are driven repeatedly in accordance with the second number of loop iterations in order to generate the second plurality of frequency ramps.

12. The system of claim 11, wherein at least one of: each frequency ramp, of the plurality of frequency ramps, is assigned to a loop iteration of the number of loop iterations, or each frequency ramp, of the second plurality of frequency ramps, is assigned to a second loop iteration of the second number of loop iterations.

13. The system of claim 11, wherein at least one of: a loop body is generated for each loop of the number of loop iterations, and the circuit elements of the transmit channel are driven repeatedly in accordance with the number of loop iterations based on the loop body, or a second loop body is generated for each loop of the second number of loop iterations, and the second circuit elements of the second transmit channel are driven repeatedly in accordance with the second number of loop iterations based on the second loop body.

14. The system of claim 11, wherein at least one of: the frequency information is updated during an execution of the number of loop iterations, or the second frequency information is updated during an execution of the second number of loop iterations.

15. The system of claim 10, wherein at least one of the first semiconductor chip or the second semiconductor chip further comprises: a receive channel for processing at least one the first radar signal or the second radar signal, wherein at least one of: the sequencing circuit is configured to drive circuit elements of the receive channel in accordance with the sequencing scheme, or the second sequencing circuit is configured to drive circuit elements of the receive channel in accordance with the second sequencing scheme.

16. The system of claim 10, wherein at least one of: the sequencing circuit is configured to use deactivation information, indicating that the transmit channel is to be deactivated, to determine the sequencing scheme, or the second sequencing circuit is configured to use second deactivation information, indicating that the second transmit channel is to be deactivated, to determine the second sequencing scheme.

17. The system of claim 10, wherein the loop instructions are executed in an instruction-processing unit with a specific instruction set for determining the sequencing scheme.

18. A method, comprising: transmitting a frequency modulated continuous wave radar signal

comprising a plurality of frequency ramps; generating, based on loop instructions, a number of loop iterations and frequency information; determining, based on the number of loop iterations and frequency information, a sequencing scheme for time-dependent functions of a transmit channel; and driving, based on the sequencing scheme, circuit components of the transmit channel in accordance with the sequencing scheme to generate the plurality of frequency ramps, wherein each frequency ramp, of the plurality of frequency ramps, is assigned to a loop iteration of the number of loop iterations.

19. A semiconductor radar chip, comprising: a transmit channel for transmitting a frequency modulated continuous wave radar signal comprising a plurality of frequency ramps; and a sequencing circuit configured to use loop instructions generating a number of loop iterations and frequency information to determine a sequencing scheme for time-dependent functions of the transmit channel and to drive circuit components of the transmit channel in accordance with the sequencing scheme, wherein the circuit components of the transmit channel are driven repeatedly in accordance with the number of loop iterations in order to generate the plurality of frequency ramps, and wherein at least one of: a loop body is generated for each loop of the number of loop iterations, wherein the circuit components of the transmit channel are driven repeatedly in accordance with the number of loop iterations based on the loop body, or the frequency information is updated during an execution of the number of loop iterations by shifting a start frequency and stop frequency.

20. The semiconductor radar chip of claim 19, further comprising: a receive channel, wherein the sequencing circuit is configured to drive circuit elements of the receive channel in accordance with the sequencing scheme.
